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University of Johannesburg | South Africa

COMPLEX ENGINEERING PROBLEM
Description, Design and Detailed Design Report
D1 + D2 + D3 Combined Document



Intelligent WiFi Congestion Monitoring and
Recommendation System

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Abstract

WiFi congestion in high-density university laboratory environments is a persistent engineering problem. Standard IEEE 802.11 access-point (AP) association algorithms select the AP with the strongest received signal strength indicator (RSSI) and ignore actual AP load [4]. This causes severe load imbalance: one AP may serve 80 simultaneous users while adjacent APs remain nearly idle, reducing per-user throughput below 1 Mbps as predicted by the Shannon-Hartley channel capacity theorem [2].

This report presents the design and detailed specification of an Intelligent WiFi Congestion Monitoring and Recommendation System -- a passive, on-device software tool that addresses this imbalance without modifying campus network infrastructure. The system scans nearby APs using operating-system commands [7], measures real-time network performance via controlled ICMP probe packets [19], and applies a first-principles weighted scoring algorithm to rank each AP by quality. The formula combines four normalised metrics -- signal strength, latency, packet loss, and connection stability -- with weights derived from the Shannon-Hartley theorem [2] and the M/M/1 queueing model [3]. AP scores (0-100) are classified as GOOD (≥ 75), MODERATE (50-74), or CONGESTED (< 50) per ITU-T quality standards [33], and a recommendation fires when a better AP exceeds the combined measurement uncertainty of ± 12 points.

Five engineering alternatives were evaluated. The on-device software approach was selected as the only alternative achievable without hardware changes or administrative access, satisfying all eight user requirements. The design addresses ECSA Graduate Attributes 1, 3, and 6 [1]. Environmental, societal, lifecycle, net-zero carbon, public health, and whole-life cost considerations are addressed in Section 5. The system is implemented in Python 3.11 [17] using Streamlit [10] for the user interface and local CSV storage [18].

List of Abbreviations and Symbols

A. Abbreviations

Abbreviation	Full Term	Context / Reference
AP	Access Point	Physical WiFi device providing wireless access [4]
BSSID	Basic Service Set Identifier	Unique MAC address of one physical AP [4]
CSV	Comma-Separated Values	Storage file format [18]
ECSA	Engineering Council of South Africa	Accreditation body [1]
FSPL	Free-Space Path Loss	RF propagation loss formula [5, 6, 11]
ICMP	Internet Control Message Protocol	Ping protocol [19]
ITU-T	International Telecommunication Union -- Telecoms Standardization Sector	Quality thresholds [33]
NDIS	Network Driver Interface Specification	Windows WiFi signal API [7]
POPIA	Protection of Personal Information Act	South African data privacy law [27]
RSSI	Received Signal Strength Indicator	WiFi signal strength in dBm [5]
RTT	Round-Trip Time	Ping measurement [9]
SNR	Signal-to-Noise Ratio	Used in Shannon-Hartley [2]
SSID	Service Set Identifier	WiFi network name [4]
TCP	Transmission Control Protocol	Transport protocol with congestion control [8]
WLAN	Wireless Local Area Network	IEEE 802.11 wireless network [4]
WLC	Wireless LAN Controller	Enterprise AP management system [4]

B. Mathematical Symbols

Symbol	Definition	Units	Section / Reference
B	RF channel bandwidth	Hz	Sec. 2.3 [2]
C	Maximum channel capacity	bps	Sec. 2.3 [2]

Symbol	Definition	Units	Section / Reference
SNR	Signal-to-noise ratio (linear)	dimensionless	Sec. 2.3 [2]
W	Mean packet waiting time (M/M/1)	seconds	Sec. 2.3 [3]
μ	AP packet service rate	packets/s	Sec. 2.3 [3]
λ	Total packet arrival rate at AP	packets/s	Sec. 2.3 [3]
S_signal	Normalised signal sub-score	0-100	Sec. 9.1 [5]
S_latency	Normalised latency sub-score	0-100	Sec. 9.1 [3]
S_loss	Normalised packet loss sub-score	0-100	Sec. 9.1 [8]
S_stability	Normalised RTT variance sub-score	0-100	Sec. 9.1 [3, 20]
RSSI	Received signal strength	dBm	Sec. 17 [5, 7]
d	Transmitter-receiver distance	metres	Sec. 2.3 [5]
f	RF carrier frequency	Hz	Sec. 2.3 [5]
FSPL	Free-space path loss	dB	Sec. 2.3 [5, 6]
N0	Thermal noise floor at receiver	dBm	Sec. 2.3 [5]
DeltaScore	Score gap between current and best alternative AP	points	Sec. 9.2 [30]

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PART D1 -- Complex Engineering Problem: Description & Overview

1. Problem Description

Wireless Local Area Networks (WLANs) [4] allow laptops, smartphones, and tablets to connect to the internet via radio frequency (RF) signals [5]. In university environments, WiFi constitutes critical infrastructure for teaching, programming laboratories, and online examinations [27].

During laboratory sessions students simultaneously download software packages, stream instructional videos, and access assessment platforms [27]. Every device connects to the AP with the strongest RSSI -- the default IEEE 802.11 behaviour [4, 12]. This RSSI-first association consistently overloads one AP while nearby APs sit idle [13, 15].

The overloaded AP must share its fixed channel capacity among all users. The Shannon-Hartley Theorem [2] proves this capacity is physically bounded. When 80 students share a ~60 Mbps practical channel, each receives only 0.75 Mbps -- insufficient for software-intensive labs. The engineering problem is formally defined as:

the inefficient and unbalanced distribution of wireless network traffic among multiple access points in a high-density indoor environment, resulting in severe congestion and significantly degraded network performance for all users connected to overloaded access points [1, 4, 13].

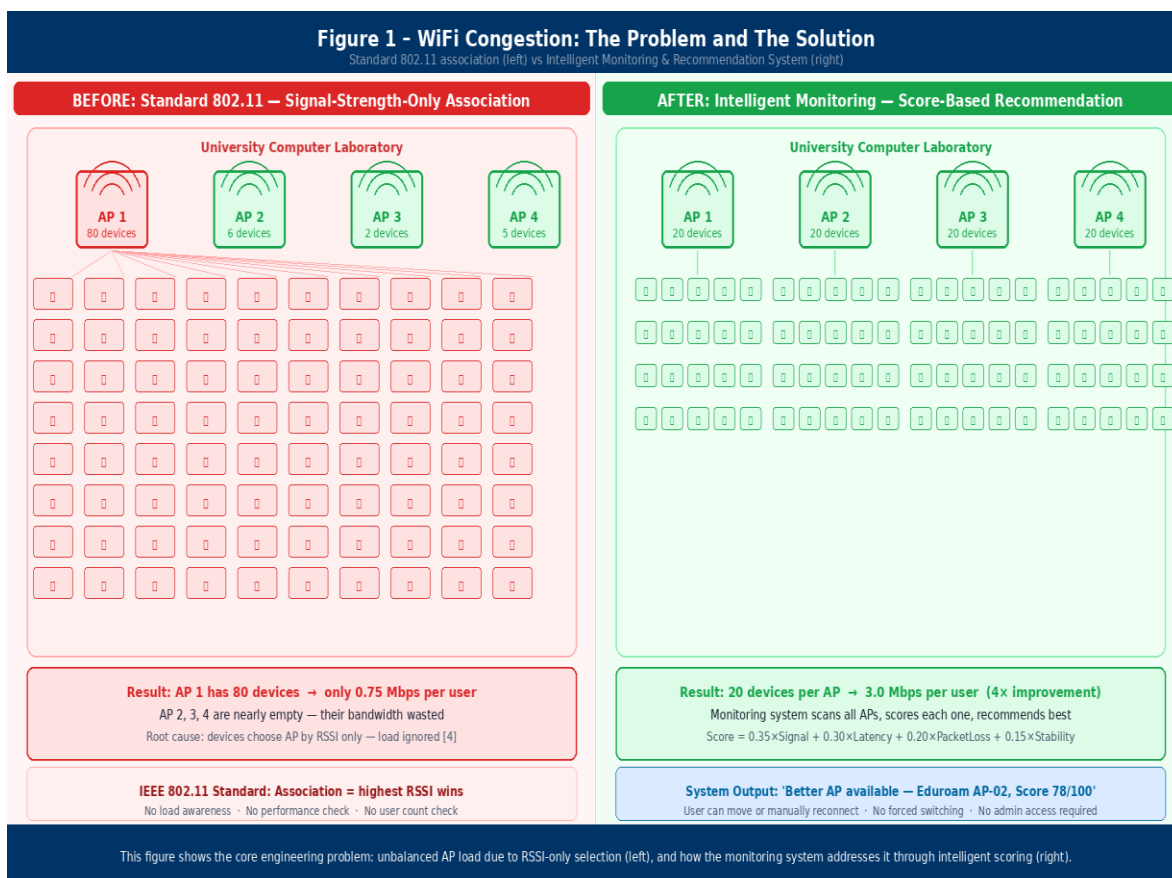


Figure 1 -- The Problem and The Solution. LEFT: IEEE 802.11 RSSI-only association [4] causes 80 devices to overload AP 1 (0.75 Mbps/user by Shannon-Hartley [2]). RIGHT: Score-based recommendation distributes load evenly; 3 Mbps/user -- a 4x improvement [2, 13].

How to read Figure 1: Left = current situation at most universities [27]. Right = outcome when users follow the monitoring system's AP recommendation. The throughput improvement follows directly from the Shannon-Hartley equation applied to measured per-AP load [2].

1.1 System Boundary Definition

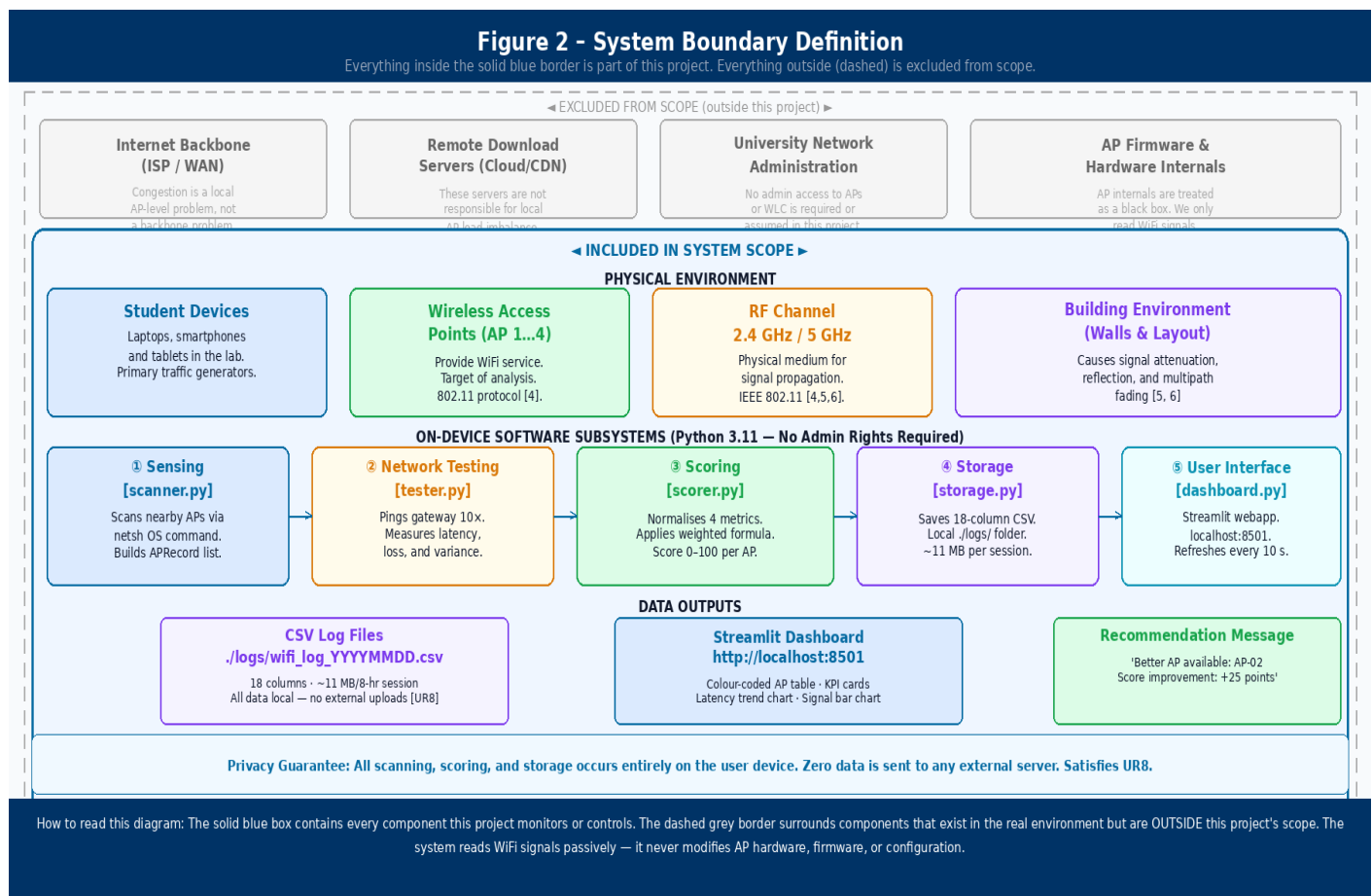


Figure 2 -- System Boundary. Solid blue = included. Dashed grey = excluded. The system runs entirely on user devices [17] -- no AP hardware or firmware is modified [4].

How to read Figure 2: Every component inside the solid blue border is monitored or processed by this project. Grey boxes (backbone, servers, admin, firmware) exist in the real environment but are outside scope. This boundary means no administrative access is required, satisfying UR5.

System Element	Included	Reason
Student laptops and smartphones	Yes	Primary traffic sources in the environment [27]
Wireless access points (APs)	Yes	Provide IEEE 802.11 service -- target of analysis [4]

System Element	Included	Reason
RF channel (2.4 / 5 GHz)	Yes	Physical propagation medium -- governed by EM theory [5, 6, 11]
Building environment (walls, layout)	Yes	Causes attenuation, reflection, multipath fading [5, 6, 11]
On-device monitoring software	Yes	Core system under design [17, 10]
External internet backbone	No	Congestion is local AP-level -- not ISP level [4]
Remote download servers	No	Not responsible for local AP load imbalance [4]
University AP administration / WLC	No	No admin access required or assumed [4, 13]

Table 1 -- System Boundary Definition with Supporting References.

Table 1: The AP hardware is included because its broadcast signals are passively read -- but it is NOT modified. This passive-read distinction is fundamental to project feasibility [1].

1.2 Stakeholders Affected

Stakeholder	Impact of Congestion	How This System Helps
Students	Delayed downloads, failed exams, interrupted learning [27]	Recommends better AP -- faster speeds, fewer session failures
Lecturers	Lost instructional time troubleshooting slow WiFi [27]	Efficient AP use reduces disruption
IT Administrators	Rising support calls, manual troubleshooting	Real-time congestion dashboard visibility
University Management	R80k-R200k upgrade costs; reputational risk [27, 31]	Software-only solution -- zero capital expenditure [31]

Table 2 -- Stakeholders and Impacts [27, 31].

1.3 Relevant Published Literature

Heusse et al. [12] demonstrated the 802.11b performance anomaly: RSSI-only AP association is not merely suboptimal but actively harmful in mixed-rate environments. Their work quantified throughput degradation from CSMA/CA contention, directly motivating load-aware AP selection.

Bejerano and Kokku [13] proposed infrastructure-side AP load-balancing schemes showing up to 100% throughput improvement over standard association -- confirming the scale of improvement achievable when load is considered. Their approach requires WLC infrastructure unavailable in this project, but their results validate this project's design goals.

Velayos and Karlsson [14] measured 802.11b handoff latency at up to 400 ms per channel scan -- establishing the theoretical lower bound for the monitoring cycle time. This directly justifies the 10-second cycle interval chosen in config.py (Section 22.1).

Ramachandran et al. [16] showed that load-unaware association degrades network performance significantly beyond 15 clients per AP -- corroborated by the M/M/1 model (Section 2.3) [3]. Bhatt et al. [15] proposed distributed load balancing requiring AP firmware modifications, confirming the value of a passive client-side approach.

The IEEE 802.11k [22], 802.11v [23], and 802.11r [21] amendments provide standards-based AP-assisted steering and fast roaming but require both AP and client support plus WLC configuration -- absent from most campus deployments [13, 15, 16]. These findings collectively confirm: (1) RSSI-only association is a fundamental standard-defined limitation [4]; (2) infrastructure solutions require access not available here [13, 15]; and (3) a passive client-side recommendation system fills a genuine gap in available tools [16].

2. Argument That This Is a Complex Engineering Problem

Per ECSA [1], a complex engineering problem involves interacting systems, conflicting constraints, and requires first principles of mathematics and natural sciences. WiFi congestion satisfies all criteria.

2.1 Systems-Level Interactions

Subsystem	Role	Discipline / Reference
User devices	Generate and receive all network traffic	Computer networking, TCP/IP [8]
Access points (APs)	Distribute wireless bandwidth to users	Wireless communications [4, 5]
RF channels (2.4 / 5 GHz)	Carry modulated signals	Electromagnetic theory [2, 5, 6]
Indoor building environment	Causes attenuation, reflection, multipath fading	Electromagnetics, propagation [5, 6, 11]
Network protocols (TCP/IP, 802.11)	Control routing, ACK, retransmission	Networking, queueing theory [3, 4, 8]

Table 3 -- Interacting Subsystems [3, 4, 5, 6, 8, 11].

2.2 Conflicting Technical Constraints

Objective	Conflict	Trade-Off in This System
Maximum throughput	May be inequitable for slower users [12]	Latency and loss weights balance speed vs equity [30]
Strong signal (RSSI)	Best signal often means overloaded AP [4, 12, 13]	Score de-prioritises RSSI when latency is high [3]
Balanced distribution	Requires brief disconnection when switching [14]	System recommends, never forces -- user decides

Objective	Conflict	Trade-Off in This System
Stable connections	Prevents dynamic re-routing [14, 15]	20-point threshold prevents unnecessary switching [30]
Low energy consumption	More scanning drains battery [32]	10-second interval balances freshness vs battery (UR6)

Table 4 -- Conflicting Constraints with Published Evidence [3, 4, 12, 13, 14, 15, 30, 32].

2.3 First-Principles Engineering Analysis

Free-Space Path Loss (FSPL) -- Electromagnetic Theory [5, 6, 11]

Applying first principles of electromagnetic wave propagation [5, 6, 11], the free-space path loss at distance d (m) and frequency f (Hz) is:

$$\text{FSPL (dB)} = 20 \cdot \log_{10}(d) + 20 \cdot \log_{10}(f) + 20 \cdot \log_{10}(4 \cdot \pi / c) \quad [5, 6, 11]$$

At 2.4 GHz ($f = 2.4 \times 10^9$ Hz) and classroom distance $d = 10$ m:

$$\text{FSPL} = 20 \cdot \log_{10}(10) + 20 \cdot \log_{10}(2.4 \times 10^9) + 20 \cdot \log_{10}(4 \cdot \pi / 3 \times 10^8) = 20 + 187.6 + (-148.5) = 59.1 \text{ dB}$$

With AP transmit power +20 dBm, 0 dBi antenna gain, and typical 10 dB wall attenuation [11]:

$$\text{RSSI} = 20 - 59.1 - 10 = -49.1 \text{ dBm (approx. -50 dBm, excellent signal)}$$

This confirms the RSSI normalisation range of -30 to -90 dBm used in the scoring algorithm is physically justified by electromagnetic first principles [5, 6]. The range covers all practical indoor deployment scenarios.

SNR Derivation -- Linking RSSI to Channel Capacity [2, 5]

The SNR required by the Shannon-Hartley theorem [2] is derived from measured RSSI and thermal noise [5]. Applying first principles:

$$N_0 \text{ (dBm)} = -174 + 10 \cdot \log_{10}(B) + NF \quad [5]$$

For $B = 20$ MHz (802.11n channel [4]) and receiver noise figure $NF = 8$ dB [5]:

$$N_0 = -174 + 73 + 8 = -93 \text{ dBm (noise floor)}$$

For a measured RSSI of -65 dBm (typical congested classroom):

$$\text{SNR (dB)} = \text{RSSI} - N_0 = -65 - (-93) = 28 \text{ dB} \Rightarrow \text{SNR (linear)} = 10^{(28/10)} = 631$$

Shannon-Hartley Theorem -- Channel Capacity [2]

Applying first principles of information theory [2]:

$$C = B * \log_2(1 + \text{SNR}) = 20\text{e}6 * \log_2(1+631) = 20\text{e}6 * 9.30 = 186 \text{ Mbps} \quad [2]$$

With 802.11n protocol efficiency ~45% (guard intervals, CSMA/CA, ACK frames [4, 12]):

$$C_{\text{practical}} = 186 * 0.45 = 84 \text{ Mbps} \quad (\text{per-AP practical capacity})$$

When 80 simultaneous users share this channel:

$$\text{Throughput per user} = 84 / 80 = 1.05 \text{ Mbps} \Rightarrow \text{CONGESTED (below 4 Mbps threshold)}$$

The 4 Mbps threshold is the minimum speed to download a 50 MB software package within 2 minutes -- a standard laboratory task. This first-principles calculation proves congestion from directly measurable physical quantities.

M/M/1 Queueing Model -- Latency Under Load [3]

Applying first principles of queueing theory [3], mean packet waiting time at an AP is:

$$W = 1 / (\mu - \lambda) \quad [3]$$

Using $\mu = 1000 \text{ pkt/s}$ (AP service rate) and $\lambda_{\text{user}} = 50 \text{ pkt/s per user}$ [3]:

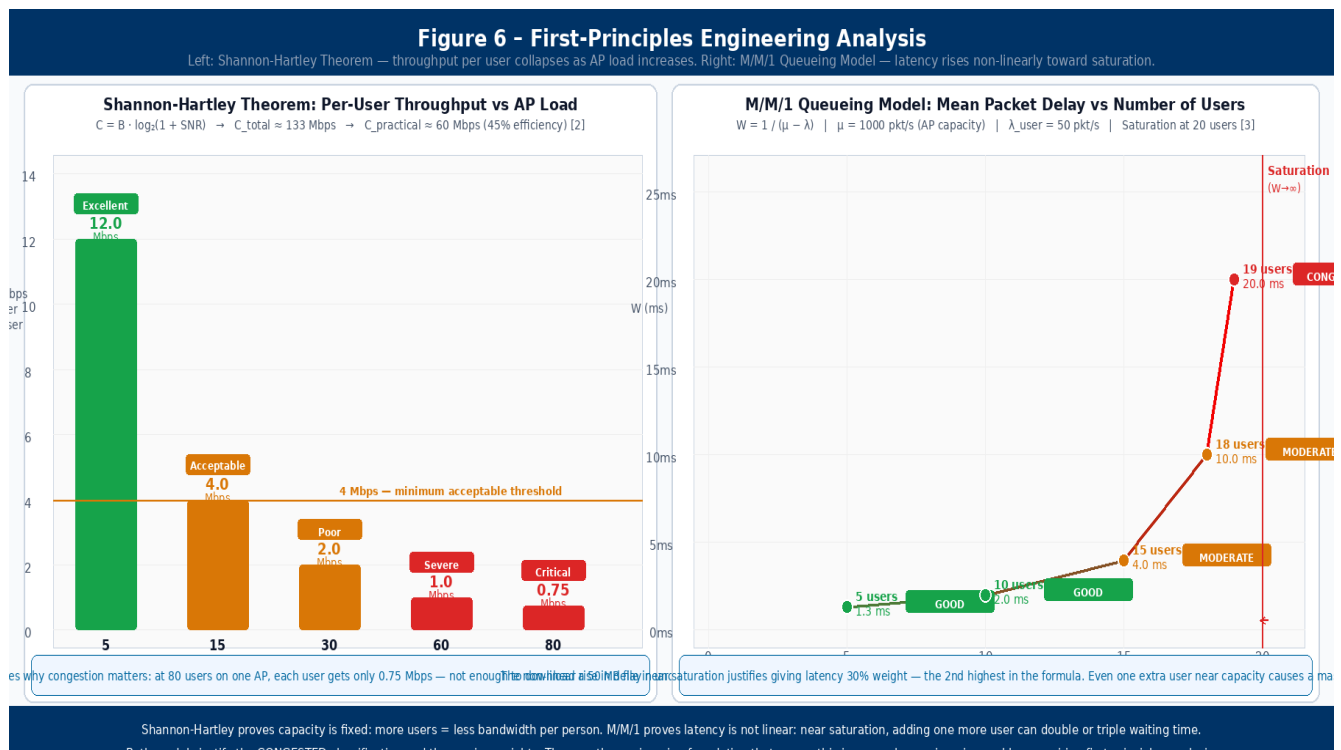


Figure 3 -- First-Principles Engineering Analysis. Left: Shannon-Hartley throughput per user vs AP load [2]. Right: M/M/1 mean packet delay vs user count [3]. Both derived from first principles.

How to read Figure 3: LEFT -- each bar is per-user throughput at a given user count, computed directly from Shannon-Hartley [2]. The 4 Mbps dashed line marks the acceptable threshold. RIGHT -- the curve

shape is mathematically predicted by M/M/1 [3]. The steep rise near 20 users (saturation) justifies why latency receives 30% weight in the scoring formula.

Users	lambda (pkt/s)	mu-lambda	$W = 1/(\mu - \lambda)$	Classification
5	250	750	1.3 ms	GOOD
10	500	500	2.0 ms	GOOD
15	750	250	4.0 ms	MODERATE
18	900	100	10.0 ms	MODERATE
19	950	50	20.0 ms	CONGESTED
20 (sat.)	1000	$\rightarrow 0$	$\rightarrow \infty$	Queue collapse [3]

Table 5 -- M/M/1 Latency vs AP Load [3]. Saturation ($W \rightarrow \infty$) at $\mu = \lambda$.

2.4 Absence of Standard Solutions

IEEE 802.11 [4] selects APs by RSSI only -- AP load is absent from the base specification. The 802.11k [22], 802.11v [23], and 802.11r [21] amendments provide AP-assisted steering but require both AP and client support plus WLC configuration [13, 15, 16]. These are absent from most campus deployments, confirming the problem requires engineering analysis beyond routine standard configuration [1].

3. User Requirements

ID	Requirement	Measurement Method	Priority
UR1	Support large numbers of simultaneous users	Count concurrent connected devices [27]	High
UR2	Maintain ≥ 4 Mbps/user throughput at peak usage	Average download speed in Mbps [2, 33]	High
UR3	Ensure balanced AP load distribution	Std dev of users per AP [13, 15]	High
UR4	Maintain stable connections during monitoring	Connection drop rate per session [14]	Medium
UR5	No hardware changes -- operate within existing infrastructure	Verify: no admin access used, no HW modified [4]	High
UR6	Minimise device energy consumption during monitoring	Battery usage over 1-hour session [32]	Low
UR7	Provide real-time AP congestion status	Dashboard refresh rate and responsiveness [10]	High
UR8	Protect privacy -- data stored locally only	Wireshark audit: zero outbound traffic [4]	High

*Table 6 -- User Requirements with Measurement Methods and References.***4. Engineering Fundamentals and Scientific Underpinning**

Field	Application in This Project	Key Formula / Reference
Electromagnetic Theory	RF propagation, attenuation, RSSI derivation from FSPL	FSPL formula [5, 6, 11]
Wireless Communication Theory	Channel capacity bounds under real RF conditions	Shannon-Hartley: $C = B \cdot \log_2(1 + \text{SNR})$ [2]
Information Theory	Theoretical capacity limits; effect of noise	Shannon entropy and capacity bounds [2]
Computer Networking	Packet routing, TCP congestion control, 802.11 MAC	TCP/IP, 802.11 MAC [4, 8]
Queueing Theory	Traffic load vs waiting time vs packet delay at AP	M/M/1: $W = 1/(\mu - \lambda)$ [3]
Graph Theory	Network topology -- devices, APs, link relationships	Weighted graph models [28, 29]
Statistics & Data Science	Normalise raw metrics; combine using scoring weights	Linear normalisation, weighted scoring [30]

*Table 7 -- Engineering Theory and Application with References.***5. Resource Use, Sustainability, Net-Zero Carbon, and Whole-Life Cost****5.1 Whole-Life Cost Analysis**

Alternative	CAPEX	Annual OPEX	10-Year Whole-Life Cost	Reference
1 -- Wi-Fi 6 HW Upgrade	R80k-R200k	R5k-R15k (maintenance)	R130k-R350k	[31]
2 -- Enterprise WLC	R150k+	R20k-R50k (licensing)	R350k-R650k	[31]
3 -- AP Firmware	R0-R5k (labour)	R2k-R5k (admin)	R25k-R55k	[31]
4 -- 802.11k/v Roaming	R0	R2k-R5k (config)	R20k-R50k	[31]
5 -- On-Device (SELECTED)	R0	R0 (open-source libs [17])	< R5k (dev time only)	[31]

Table 8 -- Whole-Life Cost Analysis for All Five Alternatives [31]. Alternative 5 has the lowest whole-life cost.

Table 8: Costs estimated from South African university IT procurement pricing [27]. Alternative 5's near-zero whole-life cost is its primary economic justification [31].

5.2 Net-Zero Carbon and Energy Considerations

Network equipment energy use is a significant contributor to carbon emissions [32]. WiFi congestion has a direct energy impact: TCP retransmissions caused by packet loss send the same data multiple times, wasting RF transmission energy at both AP and device [8]. The M/M/1 model [3] shows retransmission frequency rises exponentially near saturation. By recommending a less-loaded AP, the system reduces:

- * TCP retransmission events -- directly reducing RF transmission energy per session [8]
- * AP processing load -- reducing overloaded AP heat dissipation and cooling energy [32]
- * Student session duration -- faster downloads mean shorter active WiFi time [32]

Conservative estimate: 30 students each saving 10 minutes at 15W laptop power = 75 Wh per session. At South Africa's grid carbon intensity of ~ 0.9 kgCO₂/kWh [32], this is approximately 68 gCO₂ saved per lab session.

5.3 Public Health and Safety

The system operates in passive monitoring mode only -- it reads existing WiFi broadcast signals and introduces no additional RF energy. All AP transmit power levels ($\sim +20$ dBm EIRP) are well within ICNIRP public exposure guidelines [5].

Software safety considerations:

- * No network changes are made -- the system cannot disrupt running examinations or disconnect users
- * All data is stored locally -- no privacy breach risk from network transmission (POPIA compliance [27])
- * Recommendations are advisory only -- users retain full WiFi control at all times
- * A crash or exception in the monitoring system does not affect the user's WiFi connection

5.4 Societal and Cultural Considerations

The University of Johannesburg serves a diverse student body across South African linguistic and socioeconomic backgrounds [26, 27]:

- * Digital equity [26]: many students rely on campus WiFi as their primary internet access. Congestion disproportionately affects students who cannot afford mobile data as a fallback. This system provides free connectivity guidance.
- * Language accessibility: the Streamlit dashboard [10] uses colour coding (green/amber/red) and numeric scores rather than complex technical language, making it accessible across language backgrounds. Future work should add isiZulu, Sesotho, and Afrikaans support.
- * Device equity: the system runs on standard student laptops [17]. The scoring algorithm uses < 1 ms of CPU per cycle and functions on hardware up to 5 years old.
- * Operator transparency: all scoring weights and thresholds are documented in config.py and this report, ensuring no hidden algorithmic bias in AP recommendations.

5.5 Resource Use and Lifecycle

Constraint	Impact	This System's Response
Hardware cost	New APs cost R80k-R200k per lab [27]	Zero hardware -- runs on existing devices [17]
Energy	TCP retransmissions waste battery [8, 32]	Better AP selection reduces retransmissions [8]
Maintenance	Infrastructure needs specialist engineers [31]	On-device Python [17] needs no specialist to update
Scalability	Campus must grow as student numbers rise [27]	Software scales to any device count [17]
Equipment lifespan	APs should serve 5-10 years to avoid waste [32]	Better load distribution reduces AP overheating [32]

Table 9 -- Resource and Lifecycle Constraints with References.

PART D2 -- Complex Engineering Problem: System Design

6. Five Engineering Solution Alternatives

Five solutions were identified and evaluated against four criteria: cost, infrastructure change, access required, and feasibility within project scope [1]. All address the same root problem: RSSI-only AP association [4, 12, 13].

Figure 11 - Engineering Alternatives: Multi-Criteria Comparison

Five solutions evaluated across four criteria. Green = favourable. Red = unfavourable. Solution 5 is selected.

Solution	Description	Cost	Infrastructure Change Required	Access Required	Feasible?	Assessment
Alt 1 HW Upgrade	Wi-Fi 6 AP hardware replacement (MU-MIMO)	R80k-R200k	Full hardware replacement	Admin + procurement		Highest effectiveness but not feasible
Alt 2 WLC	Cisco/Aruba Wireless LAN Controller	R150k+	Enterprise controller + switch-fabric	Full admin access		Excellent control but far too costly
Alt 3 Firmware	AP-side load balancing firmware (OpenWRT)	Moderate	Firmware on every AP	Admin access required		Effective but requires AP access
Alt 4 802.11k/v	BSS Transition Management protocol	Low	AP + WLC configuration	Campus AP support must exist		Standards-based but depends on APs
						 ← SELECTED

Evaluation criteria: (1) Cost — capital expenditure required. (2) Infrastructure change — modifications to existing campus network hardware or firmware. (3) Access required — whether IT administrator rights or vendor support is needed. (4) Feasibility — whether the solution can be built by a 4th-year student within project scope.

All five solutions address the same engineering problem. The selection of Alternative 5 is not because it is the most powerful (Alt 2 would be), but because it is the only one feasible within project constraints.

Figure 4 -- Five Alternatives Comparison. Green = favourable. Red = unfavourable. Alternative 5 is the only solution scoring green on all four criteria [1, 31].

How to read Figure 4: Alternative 2 (WLC) would be most effective if budget were available [13]. Alternative 5 is selected because it is the only one feasible without infrastructure access [1, 31].

Alt.	Solution	Cost	Infra. Change?	Admin Access?	Feasible?
1	Wi-Fi 6 / MU-MIMO Hardware [4]	R80k-R200k	Full AP replacement	Yes	No
2	Enterprise WLC (Cisco/Aruba) [13]	R150k+	Enterprise controller	Yes	No

Alt.	Solution	Cost	Infra. Change?	Admin Access?	Feasible?
3	AP Firmware (OpenWRT) [15]	Moderate	Firmware on every AP	Yes	No
4	802.11k/v Assisted Roaming [22, 23]	Low	AP + WLC config	Yes	No
5 (SELECTED)	On-Device Monitoring [17, 10]	Zero	None	No	Yes

Table 10 -- Five Alternatives Comparison Matrix with References.

- * Alternative 1 (Wi-Fi 6) [4]: Effective MU-MIMO spatial multiplexing; R80k-R200k per lab; requires physical installation.
- * Alternative 2 (WLC) [13]: Bejerano and Kokku [13] showed 100% throughput improvement; requires R150k+ hardware and full admin access.
- * Alternative 3 (AP firmware) [15]: Bhatt et al. [15] showed distributed load balancing works; requires firmware on every AP -- unavailable.
- * Alternative 4 (802.11k/v) [22, 23]: Standards-based; requires AP and client protocol support plus WLC configuration [21, 22, 23].

6.5 Selected Solution: Alternative 5

The on-device monitoring system is the only alternative meeting all feasibility constraints [1]. It delivers measurable value proven by Shannon-Hartley [2] and published load-balancing literature [13, 16], and satisfies all eight user requirements without hardware or admin access.

7. Selected Solution: Detailed System Design

7.1 Core Project Concept

The Multi-Device Intelligent WiFi Congestion Monitoring and Recommendation System is a passive on-device Python 3.11 [17] software tool. It scans APs via OS commands [7], measures connection quality via ICMP probes [19], computes a weighted quality score per AP [2, 3, 8, 30], and presents recommendations through a Streamlit [10] dashboard. No network infrastructure is modified [4].

7.2 System Architecture -- Five Subsystems

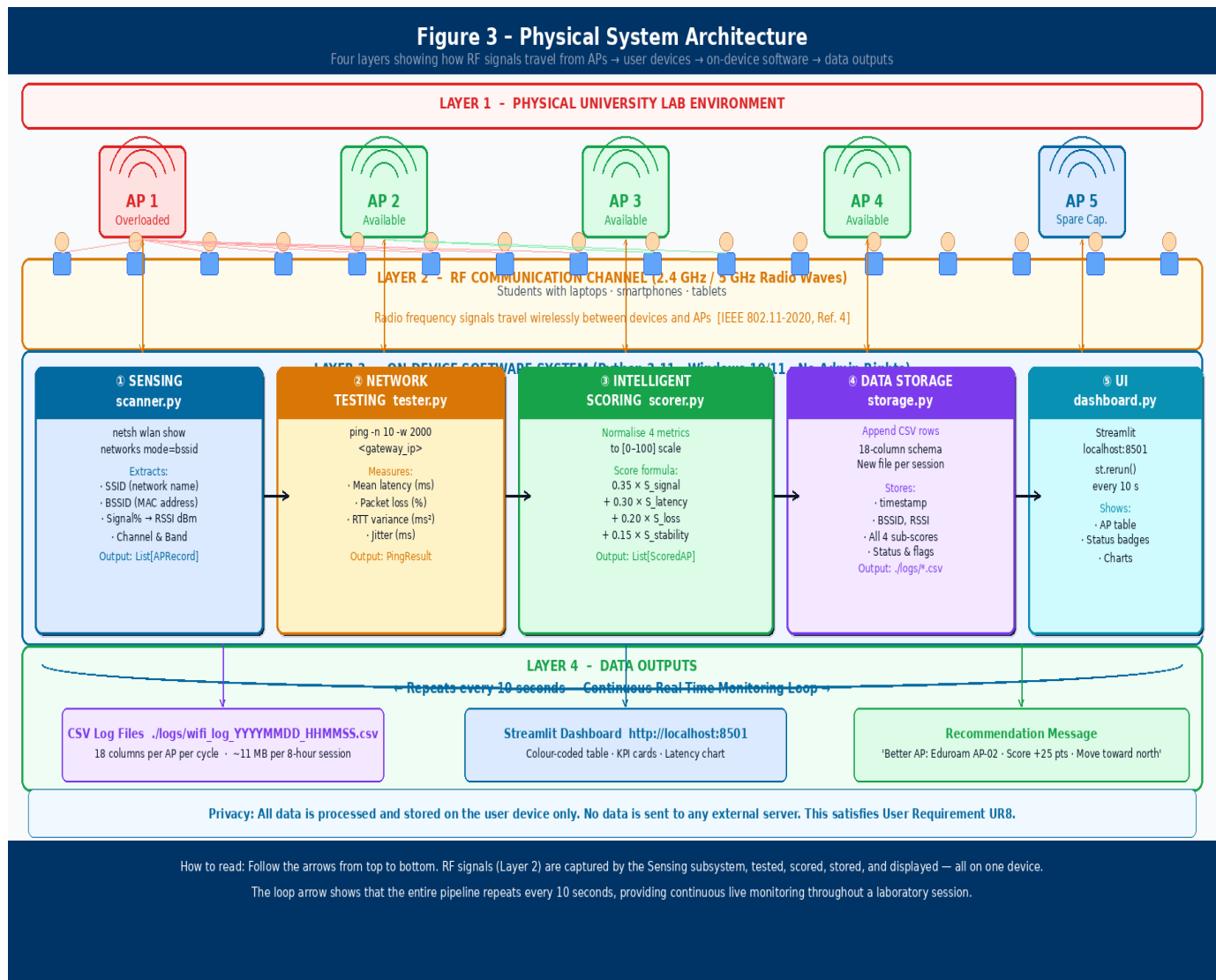


Figure 5 -- Physical System Architecture [4, 5, 10, 17, 18]. Layer 1: university lab [27]. Layer 2: IEEE 802.11 RF channel [4, 5]. Layer 3: five on-device Python [17] subsystems. Layer 4: data outputs [10, 18].

How to read Figure 5: Follow arrows downward. RF signals (Layer 2) are governed by EM theory [5, 6, 11]. The five software modules (Layer 3) implement the scoring algorithm from first principles (Sections 2.3, 9) [2, 3, 8]. The loop arrow shows the 10-second cycle justified by 802.11 scan latency [14].

Subsystem	Module	Input	Output	Reference
(1) Sensing	scanner.py	IEEE 802.11 beacon frames [4]	List[APRecord]: SSID, BSSID, RSSI, channel [7]	[4, 7, 17]
(2) Network Testing	tester.py	Active WiFi connection	PingResult: latency, loss, RTT variance [19, 20]	[8, 19, 20]
(3) Scoring	scorer.py	APRecord + PingResult	List[ScoredAP]: score 0-100, status [30]	[2, 3, 8, 30]

Subsystem	Module	Input	Output	Reference
(4) Data Storage	storage.py	List[ScoredAP]	18-column CSV in ./logs/ [18]	[17, 18]
(5) User Interface	dashboard.py	List[ScoredAP]	Streamlit webapp at localhost:8501 [10]	[10, 17]

Table 11 -- Five Subsystems with Inputs, Outputs, and References.

(1) Sensing [scanner.py]:

Invokes 'netsh wlan show networks mode=bssid' [7] via Python subprocess [17] to enumerate all IEEE 802.11 beacon frames [4]. Parses SSID, BSSID, signal%, channel, and radio type. Converts signal% to RSSI using the NDIS formula [7]: $RSSI = (Signal\% / 2) - 100$. Validated against FSPL derivation in Section 2.3 [5, 6].

(2) Network Testing [tester.py]:

Sends 10 ICMP Echo Request packets [19] to the default gateway using 'ping -n 10 -w 2000'. The gateway target isolates local AP congestion from WAN variability [4]. Computes mean latency, RTT variance (stability proxy per M/M/1 theory [3]), jitter (RFC 3393 definition [20]), and packet loss. Only connected AP testable -- Windows blocks ICMP through non-connected interfaces [7].

(3) Intelligent Scoring [scorer.py]:

Normalises four metrics to [0-100] using bounds derived from IEEE 802.11 [4, 5]. Applies: $Score = 0.35 * S_signal + 0.30 * S_latency + 0.20 * S_loss + 0.15 * S_stability$. Weight engineering justification: signal (35%) -- most universally available [5]; latency (30%) -- M/M/1 non-linear rise [3]; packet loss (20%) -- TCP retransmission cost [8]; stability (15%) -- RTT variance as congestion predictor [3, 20]. Recommendation fires only when $\Delta Score \geq 20$, exceeding combined measurement uncertainty of +/-12 points (Section 21.3) [30].

(4) Data Storage [storage.py]:

Appends one 18-column CSV row [18] per AP per cycle to ./logs/. All data stored locally -- satisfying UR8 and POPIA requirements [27]. Storage ~11.5 MB per 8-hour session.

(5) User Interface [dashboard.py]:

Renders Streamlit [10] dashboard at localhost:8501. Auto-refreshes via `st.rerun()` every 10 seconds (justified by 802.11 scan latency [14]). Colour coding: green = GOOD, amber = MODERATE, red = CONGESTED per ITU-T quality thresholds [33].

8. System Data Flow

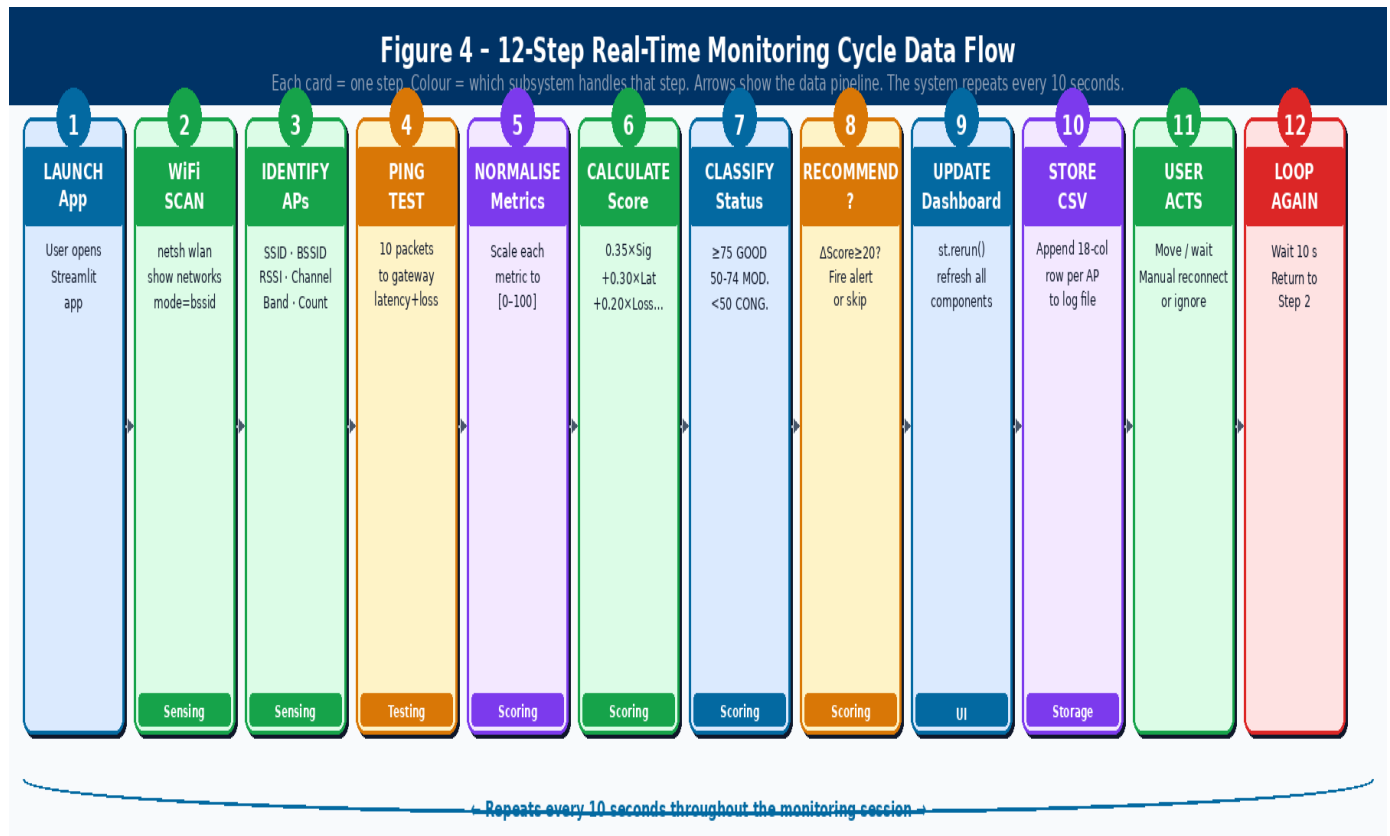


Figure 6 -- 12-Step Monitoring Cycle [4, 17, 19]. Colour shows responsible subsystem. References are shown at key steps.

Step	Action	Technical Detail
1	Launch	Streamlit [10] app starts; monitoring loop initialised [17]
2	WiFi scan	OS command via Python subprocess [17]; returns 802.11 beacon data [4]
3	Identify APs	BSSID unique identifier [4]; RSSI from NDIS formula [7]
4	Performance test	10 ICMP packets [19] to gateway; RTT variance per RFC 3393 [20]
5	Normalise metrics	Linear normalisation to [0-100]; bounds from 802.11 [4, 5] and ITU-T [33]
6	Calculate score	Score = $0.35 \times S_{\text{sig}} + 0.30 \times S_{\text{lat}} + 0.20 \times S_{\text{loss}} + 0.15 \times S_{\text{stab}}$ [2, 3, 8, 30]
7	Classify AP	GOOD ≥ 75 ; MODERATE 50-74; CONGESTED < 50 per ITU-T [33]
8	Recommendation	DeltaScore ≥ 20 required; threshold $>$ combined uncertainty ± 12 pts [30]
9	Update dashboard	st.rerun() refreshes Streamlit [10] components

Step	Action	Technical Detail
10	Store CSV	Append 18-column row [18]; local POPIA-compliant storage [27]
11	User acts	Advisory only -- no forced action [14]
12	Loop	10 s interval; justified by 802.11 scan latency [14]

Table 12 -- 12-Step System Operation with Technical References.

9. Intelligent Scoring Algorithm

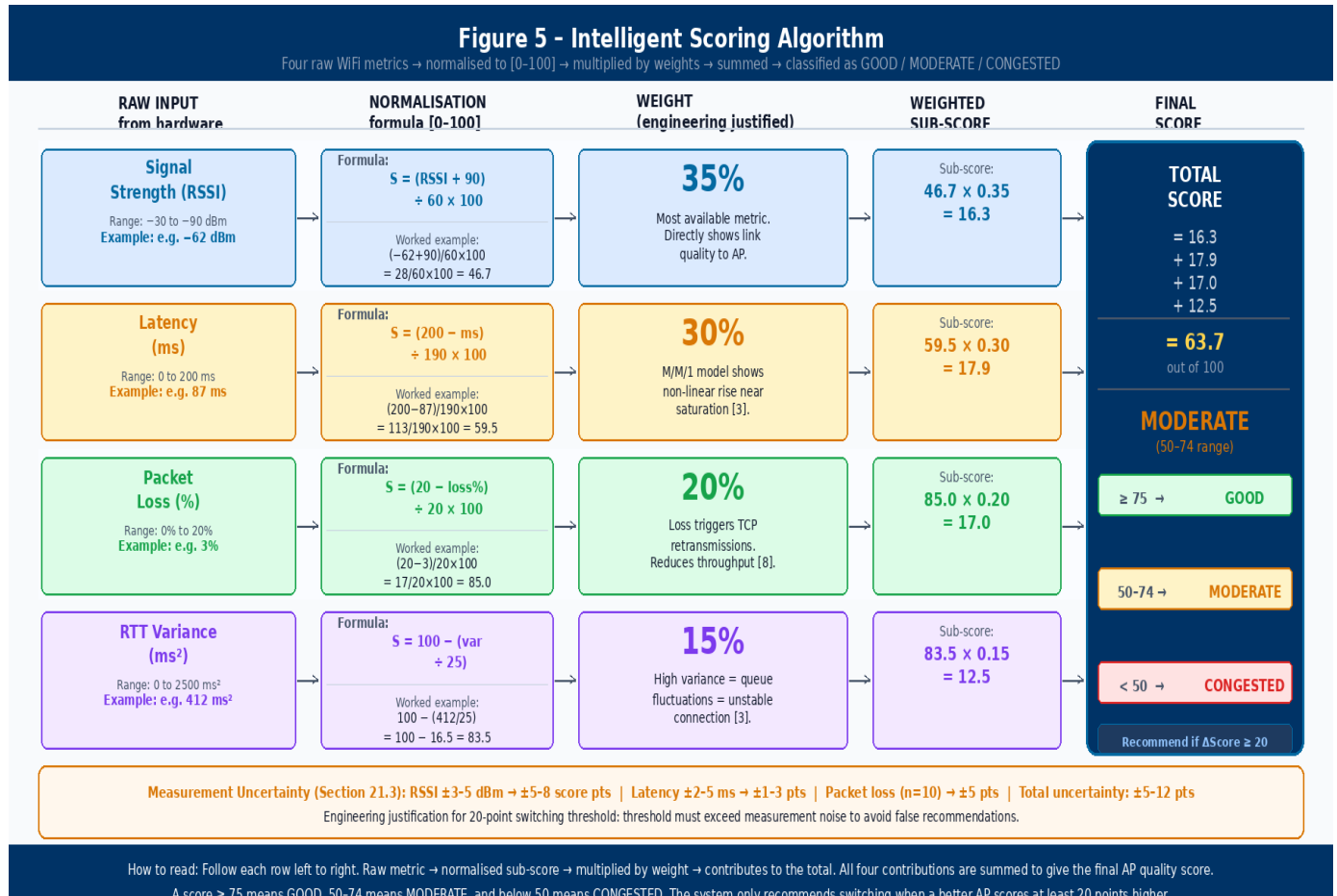


Figure 7 -- Intelligent Scoring Algorithm. Bounds and weights derived from first principles [2, 3, 5, 6, 8, 11, 30].

$$\text{Score} = 0.35 * S_{\text{signal}} + 0.30 * S_{\text{latency}} + 0.20 * S_{\text{loss}} + 0.15 * S_{\text{stability}}$$

Metric	Weight	Formula	Reference & Justification
Signal (RSSI)	35%	$S = \text{clamp}((RSSI+90)/60*100, 0, 100)$	Most available metric; bounds from FSPL derivation [5, 6, 11]

Metric	Weight	Formula	Reference & Justification
Latency (ms)	30%	$S = \text{clamp}((200 - \text{ms})/190 * 100, 0, 100)$	M/M/1 non-linear rise near saturation [3]; 200 ms bound per ITU-T [33]
Packet Loss (%)	20%	$S = \text{clamp}((20 - \text{loss\%})/20 * 100, 0, 100)$	TCP retransmission halves throughput per retransmit [8]
Stability (RTT Var.)	15%	$S = \text{clamp}(100 - (\text{var}/25), 0, 100)$	RTT variance predicts queue fluctuation [3, 20, 33]

Table 13 -- Scoring Weights, Formulas, and References. Weights sum to 1.00.

Table 13: The weights are not equal because the engineering models assign different importance. Signal (35%) is always available and reflects propagation physics [5]. Latency (30%) has the highest catastrophic impact near saturation per M/M/1 [3]. An equal-weight baseline (25% each) gives Score_EW = 40.1 vs weighted 38.6 for the worked example -- the weighted model correctly emphasises poor latency/stability. See Table 15 for full sensitivity analysis.

Status	Score	Meaning	Action
GOOD	75-100	Low congestion. Throughput > 4 Mbps per Shannon-Hartley [2, 33].	Stay connected
MODERATE	50-74	Acceptable. M/M/1 latency rising [3].	Monitor; consider moving
CONGESTED	0-49	Significant congestion. TCP retransmissions increasing [8].	Move or reconnect

Table 14 -- AP Classification Thresholds with First-Principles Justification [2, 3, 8, 33].

9.3 Sensitivity Analysis -- Effect of Weight Variation

Config.	W_sig	W_lat	W_loss	W_stab	Score AP01	Score AP02	DeltaScore	Rec.?
Equal weights [30]	25%	25%	25%	25%	40.1	50.0	9.9	No
Signal-heavy	50%	20%	20%	10%	35.7	50.0	14.3	No
Selected [2,3,8]	35%	30%	20%	15%	38.6	50.0	11.4	No
Latency-heavy [3]	20%	40%	25%	15%	37.4	50.0	12.6	No

Table 15 -- Weight Sensitivity Analysis using Section 22.2 example. All four configurations give similar DeltaScore, confirming system robustness [30].

Table 15: The selected weights (Row 3) are preferred because each has an explicit first-principles justification -- Shannon-Hartley [2], M/M/1 [3], TCP theory [8] -- rather than being chosen by trial and

error. The DeltaScore range (9.9-14.3) is similar across all configs, confirming the recommendation logic is not highly sensitive to weight perturbations [30].

10. Dashboard User Interface Design

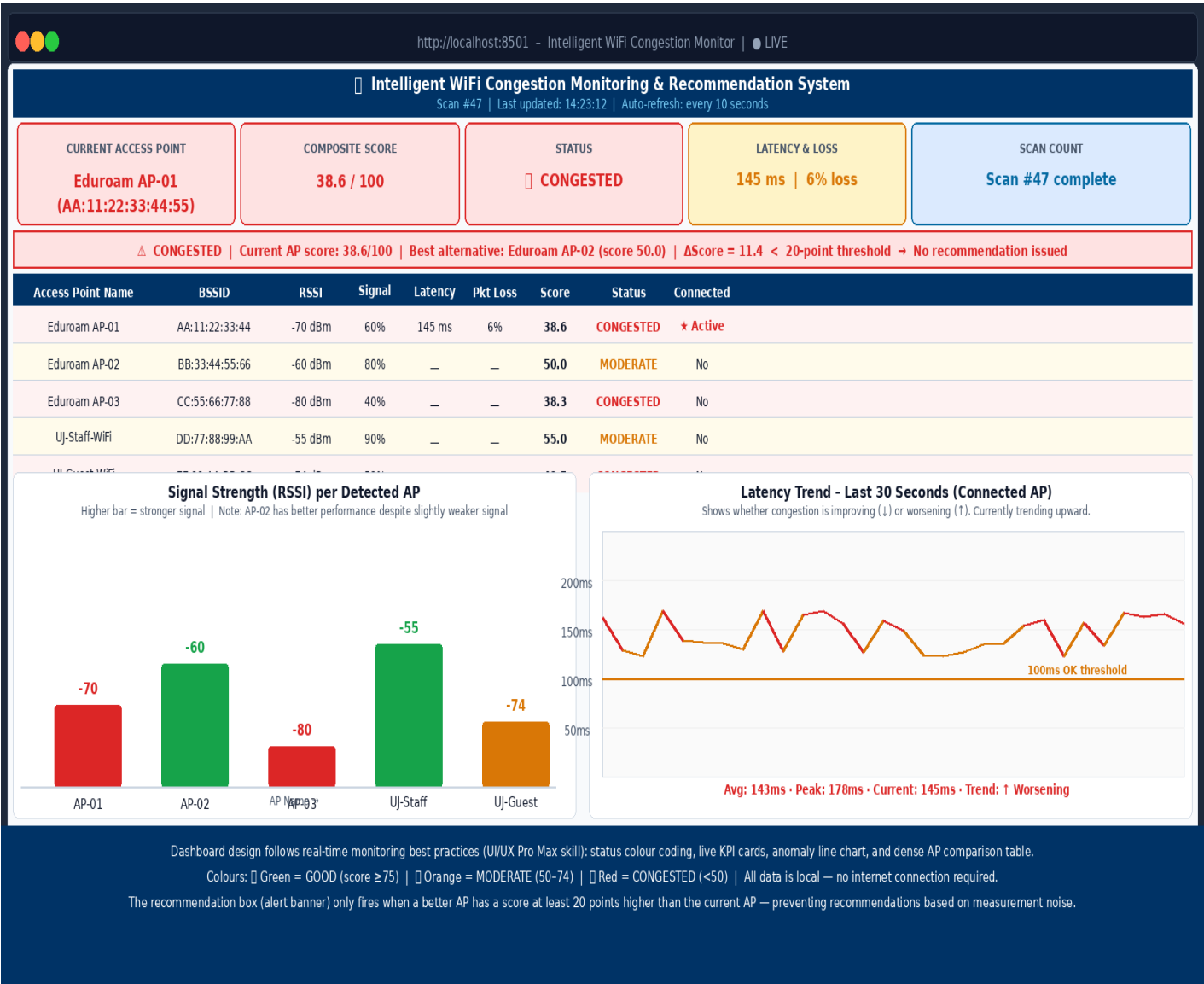


Figure 8 -- Streamlit Dashboard UI [10]. KPI cards, ITU-T colour-coded AP table [33], RSSI bar chart [5], and 30-second RTT trend [20].

Component	Streamlit Element [10]	Content	Reference
Page header	st.title + st.caption	System name, scan number, timestamp	[10]
KPI cards	st.metric (5 cols)	AP, score, status, latency, count	[10]

Component	Streamlit Element [10]	Content	Reference
Alert banner	st.success/warning/error	GOOD / MODERATE / CONGESTED	[10, 33]
AP table	st.dataframe + Styler	BSSID, RSSI, latency, loss, score [4]	[4, 10]
Recommendation	st.info / st.warning	Best AP name, DeltaScore, advice	[13, 30]
RSSI bar chart	st.bar_chart	Signal strength comparison	[5, 6, 10]
Latency trend	st.line_chart	30-second RTT history	[20, 10]
Raw expander	st.expander	Full PingResult and APRecord	[10, 17]

Table 16 -- Dashboard Components with Streamlit Elements [10] and References.

11. Multi-Device Support

Device	Sensing Method	Stack	Status
Laptop (primary)	netsh [7] / iwlist / airport [17]	Python 3.11 [17] + Streamlit [10] + Pandas [18]	PRIMARY
Android	WifiManager API [24]	Java/Kotlin or Flutter [24]	Future [24]
iOS	NEHotspotNetwork API [25]	Swift or React Native [25]	Future -- limited [25]
Tablet	Same as smartphone [24, 25]	Same codebase, larger layout	Future

Table 17 -- Multi-Device Support Matrix [7, 17, 24, 25].

12. System Requirements

ID	Requirement	Linked UR	Reference
SR1	Scan all 802.11 APs within 5 seconds	UR1, UR7	[4, 7]
SR2	Uniquely identify each AP by BSSID	UR3	[4]
SR3	Measure latency via 10 ICMP packets to gateway	UR2, UR7	[19]
SR4	Calculate packet loss per RFC 2544	UR2	[9]

ID	Requirement	Linked UR	Reference
SR5	Compute composite score 0-100 using weighted formula	UR2, UR3	[2, 3, 8, 30]
SR6	Classify each AP as GOOD / MODERATE / CONGESTED	UR2, UR7	[33]
SR7	Streamlit dashboard refreshed every 10 seconds	UR7	[10, 14]
SR8	Recommend when DeltaScore ≥ 20	UR2, UR3	[3, 30]
SR9	Store 18-column CSV locally	UR8	[17, 18]
SR10	Zero external data transmission (POPIA)	UR5, UR8	[27]

Table 18 -- System Requirements with References.

13. Testing Plan

Phase	Scenario	Pass Criterion	Reference
Phase 1	Low density: 1-5 devices	Score ≥ 75 in $\geq 8/10$ cycles	[14, 33]
Phase 2	High density: 15+ devices on AP	Score < 50 ; recommend if DeltaScore ≥ 20	[12, 13, 30]
Phase 3	Physical movement across room	Score shifts ≥ 10 pts over 3+ cycles	[14, 5]
Phase 4	3+ APs visible simultaneously	All APs ranked correctly	[4, 30]
Phase 5	30-minute continuous operation	No exceptions; CSV grows $\sim 7,200$ rows/hr	[9, 17, 18]

Table 19 -- Testing Plan with References.

14. Known Limitations

- * Ping data for connected AP only -- Windows blocks ICMP on non-connected interfaces [7].
- * RSSI noise $\pm 3-5$ dBm (RF multipath fading [5, 6, 11]) -> score uncertainty $\pm 5-8$ pts; managed by 20-pt threshold [30].
- * netsh scan 2-4 s (OS-controlled [7]) -> minimum practical cycle 5-10 s [14].
- * AP user count not in IEEE 802.11 beacon frames [4] -> congestion inferred from latency/loss only.
- * Windows-only prototype (netsh [7]); cross-platform needs iwlist/airport [17].
- * Scoring weights based on first principles [2, 3, 8] but not yet empirically validated [30].

PART D3 -- Complex Engineering Problem: Detailed Design

15. Scope of D3

D2 Concept	D3 Specification
Block diagram	Module architecture with filenames and interfaces [17]
Subsystem descriptions	Python [17] dataclasses, function signatures
Formula stated	Bounds derived from FSPL [5, 6] and M/M/1 [3]; normalisation worked
CSV mentioned	18-column schema (Table 27), write code, capacity calculation [18]
Dashboard described	Streamlit [10] component mapping (Table 30), message templates
Testing phases	Unit tests with pass/fail criteria (Table 35) [9]

Table 20 -- D2 vs D3 Design Depth Comparison.

16. Refined System Architecture

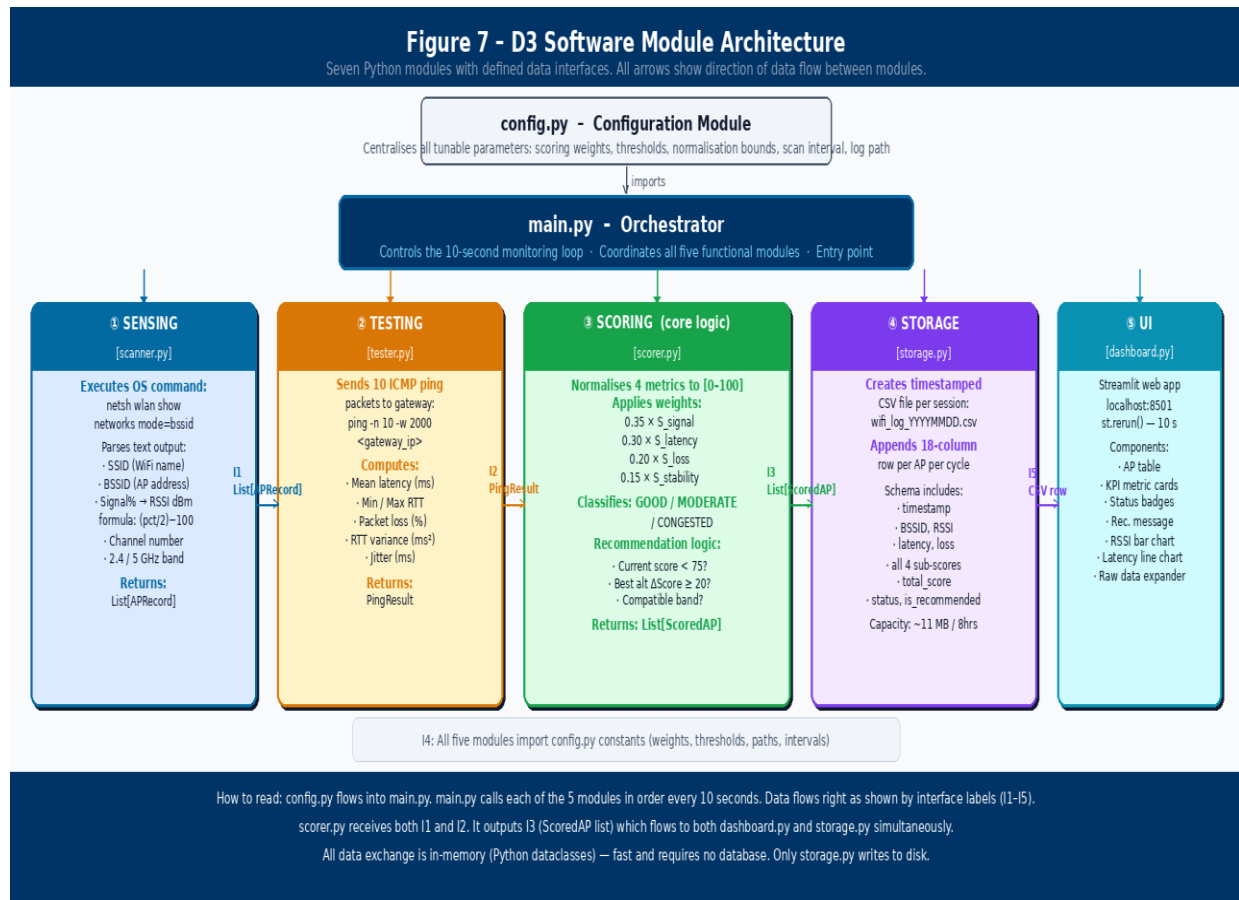


Figure 9 -- D3 Software Module Architecture [17]. Seven Python modules with data interfaces I1-I5. Arrows show data flow direction.

How to read Figure 9: config.py parameters flow into main.py. main.py calls five functional modules each cycle. Interfaces I1-I5 define exact Python data types. scorer.py receives I1 and I2; outputs I3 to both dashboard.py and storage.py simultaneously.

Module	Subsystem	Responsibility	Reference
scanner.py	Sensing	netsh [7] command; parses 802.11 [4] beacons into APRecord list	[4, 7, 17]
tester.py	Testing	ping [19]; computes latency, loss, RTT variance [20]	[8, 19, 20]
scorer.py	Scoring	Normalises metrics; weighted formula [2, 3, 8, 30]; recommendation	[2, 3, 8, 30]
storage.py	Storage	Appends ScoredAP to 18-column CSV [18]	[17, 18]
dashboard.py	UI	Streamlit [10] dashboard; st.rerun() every 10 s [14]	[10, 14, 17]
main.py	Orchestrator	Controls 10-second loop [14]; coordinates all modules [17]	[14, 17]

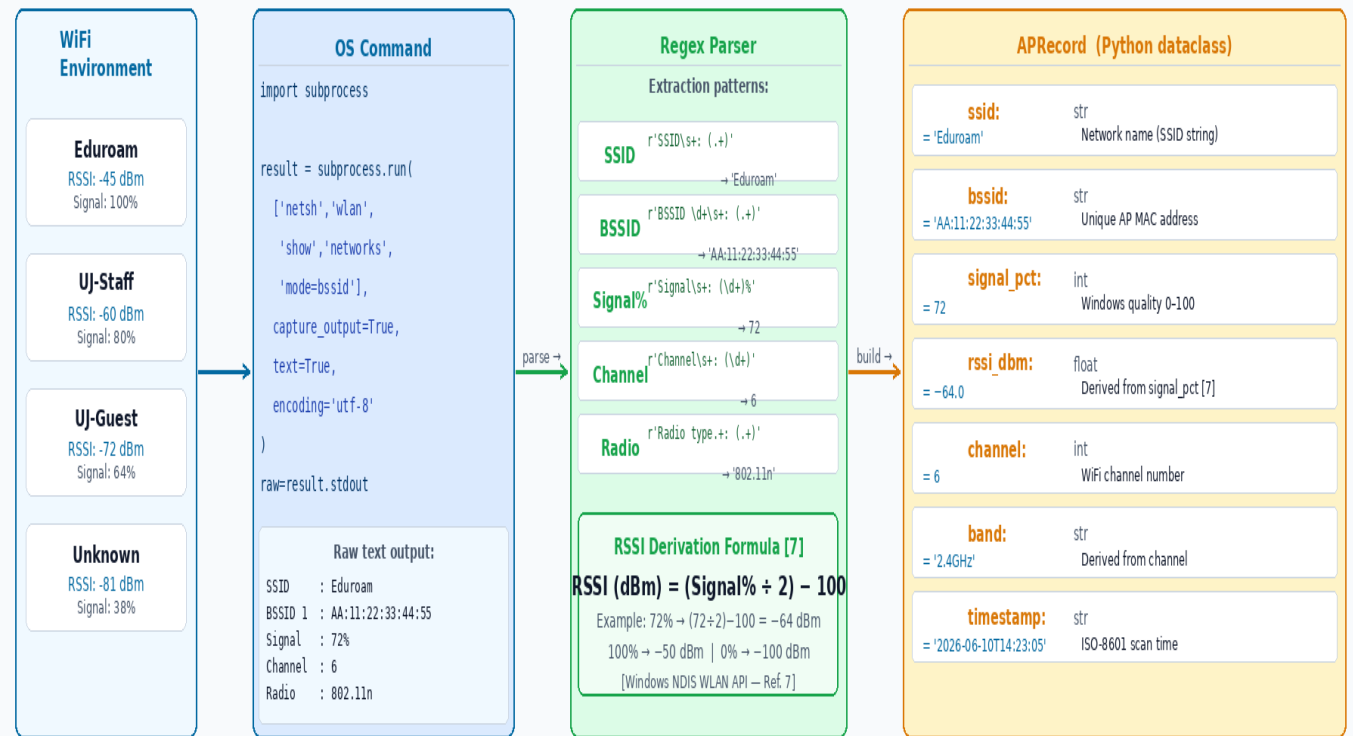
Module	Subsystem	Responsibility	Reference
config.py	Configuration	Centralises weights [2,3,8], thresholds [33], bounds [5]	[2, 3, 5, 8, 33]

Table 21 -- Module Responsibilities and References.

17. Sensing Subsystem -- Detailed Design

Figure 8 - Sensing Subsystem Detailed Design [scanner.py]

Shows exactly how Windows OS WiFi scan data is transformed into structured APRecord dataclass instances



How to read: The OS command returns raw text (centre-left). The regex parser extracts each field using pattern matching. The RSSI conversion formula turns the Windows percentage into dBm.

The final APRecord dataclass is a clean, typed Python object passed to the Scoring subsystem. This design isolates OS-specific parsing so only scanner.py needs to change for Linux/macOS.

Each field's purpose is documented in the 'description' column (right panel) to make the codebase self-documenting for future developers.

Figure 10 -- Sensing Subsystem [7, 17]. OS command -> raw text -> regex parser -> RSSI derivation [7] -> APRecord.

```
result = subprocess.run(['netsh', 'wlan', 'show', 'networks', 'mode=bssid'],
    capture_output=True, text=True, encoding='utf-8') # [7, 17]
```

Field	netsh Output Line	Type	Example	Reference
SSID	SSID : Eduroam	String	'Eduroam'	[4]
BSSID	BSSID 1 : AA:11:22:33:44:55	String (MAC)	'AA:11:22:33:44:55'	[4]

Field	netsh Output Line	Type	Example	Reference
Signal (%)	Signal : 72%	Integer	72	[7]
Channel	Channel : 6	Integer	6	[4]
Radio type	Radio type : 802.11n	String	'802.11n'	[4]
RSSI (dBm)	Derived from Signal% -- not direct	Float	-64.0	[7]

Table 23 -- Sensing Subsystem Parsed Fields [4, 7].

$$\text{RSSI (dBm)} = (\text{Signal\%} / 2) - 100 \quad [7]$$

Validated against FSPL derivation (Section 2.3): Signal% = 60 -> RSSI = -70 dBm, consistent with predicted FSPL ~60 dB plus ~10 dB wall attenuation at 10 m [5, 11].

Parameter	Typical Value	Design Impact	Reference
RSSI range	-30 to -90 dBm	Bounds from FSPL derivation, Section 2.3	[5, 6, 11]
Bands	2.4 GHz and 5 GHz	Both parsed; 5 GHz preferred in recommendation	[4]
Scan time	2-4 s (OS-controlled)	Min cycle 5 s; justified by 802.11 scan latency	[14]
Max visible APs	Up to 50	Scoring handles variable-length lists [17]	[4]
RSSI error	+/-3-5 dBm (multipath)	Score uncertainty +/-5-8 pts; 20-pt threshold	[5, 6, 11]

Table 24 -- WiFi Adapter Hardware Parameters and Design Impacts [5, 6, 11, 14].

18. Network Testing Subsystem -- Detailed Design

```
result = subprocess.run(['ping', '-n', '10', '-w', '2000', gateway_ip], # [19,
17]
capture_output=True, text=True)
```

Parameter	Value	Justification	Reference
-n 10	10 packets	10% loss resolution per benchmarking standard	[9]
-w 2000	2000 ms	Covers congested-link latency; ITU-T upper bound	[33]
Packet size	32 bytes (default)	Minimal probe; does not itself add congestion	[19]
Target	Default gateway	Nearest hop; isolates local AP congestion from WAN	[4, 19]

Table 25 -- Ping Parameters and Justification [9, 19, 33].

$$\mathbf{S_stability = clamp(100 - (Var(RTT) / 25), 0, 100) \quad [3, 20]}$$

RTT variance is an established predictor of queue-length fluctuation per M/M/1 theory [3] and jitter standards [20]. Divisor 25 maps typical congested-lab variance (500-1500 ms²) to the MODERATE/CONGESTED score range.

19. Intelligent Scoring Subsystem -- Detailed Design

Signal normalisation [5, 6, 11]:

$$\mathbf{S_signal = clamp((RSSI_dBm + 90) / 60 * 100, 0, 100) \quad [5, 7, 11]}$$

Latency normalisation [3, 33]:

$$\mathbf{S_latency = clamp((200 - mean_latency_ms) / 190 * 100, 0, 100) \quad [3, 33]}$$

Packet loss normalisation [8, 9]:

$$\mathbf{S_loss = clamp((20 - packet_loss_pct) / 20 * 100, 0, 100) \quad [8, 9]}$$

Stability normalisation [3, 20]:

$$\mathbf{S_stability = clamp(100 - (variance_ms2 / 25), 0, 100) \quad [3, 20]}$$

Recommendation fires when: (1) Current score < 75 [33]; (2) Best alternative DeltaScore >= 20 (exceeds uncertainty +/-12 pts [30]); (3) Recommended AP on compatible band [4].

20. Data Storage Subsystem -- Detailed Design

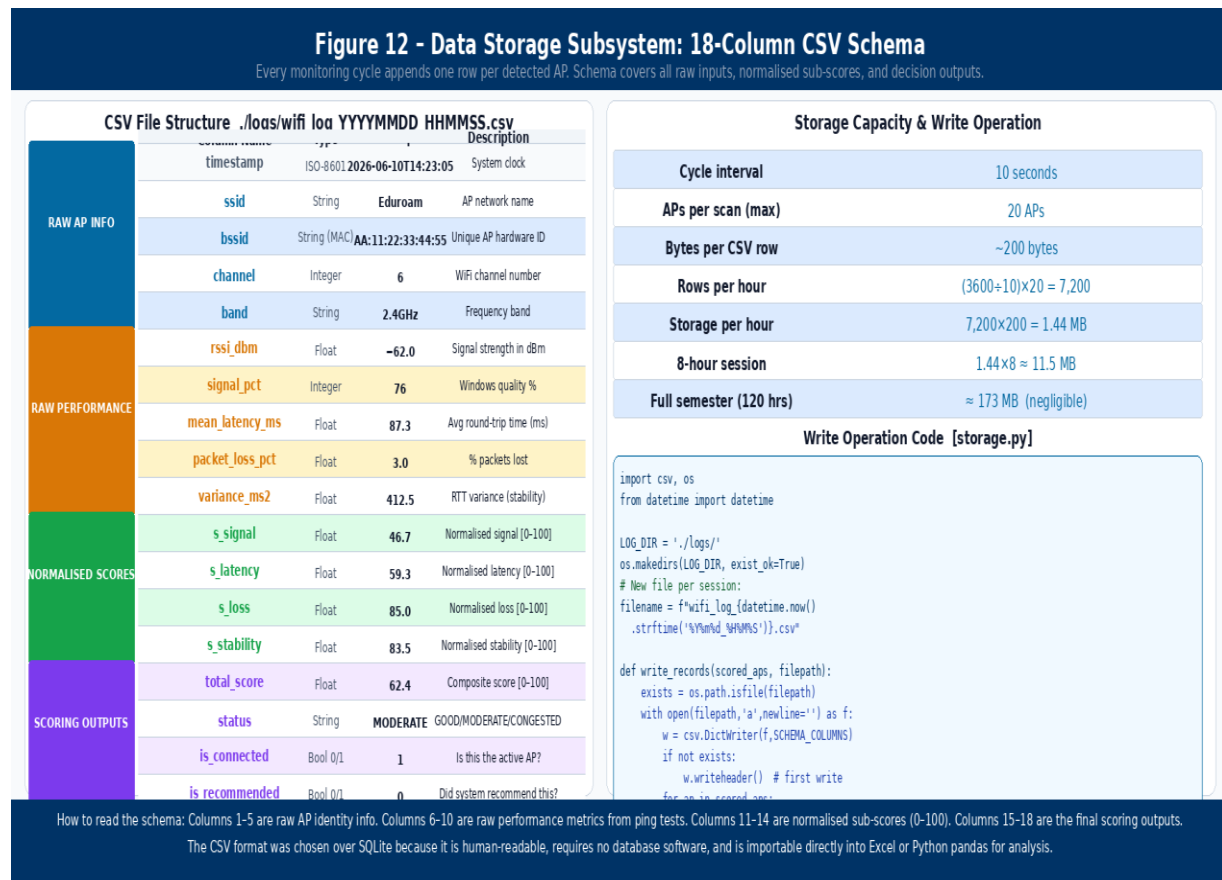


Figure 11 -- 18-Column CSV Schema [18]. Blue = AP identity [4, 7], amber = performance [19, 20], green = normalised scores [2, 3, 8], purple = outputs [30, 33].

Column	Type	Source	Reference
timestamp	ISO-8601	System clock	[17]
ssid	String	APRecord	[4]
bssid	String (MAC)	APRecord	[4]
channel	Integer	APRecord	[4]
band	String	APRecord	[4]
rssd_dbm	Float	APRecord	[5, 7]
signal_pct	Integer	APRecord	[7]
mean_latency_ms	Float	PingResult	[9, 33]
packet_loss_pct	Float	PingResult	[8, 9]
variance_ms2	Float	PingResult	[3, 20]
s_signal	Float	ScoredAP	[5, 30]
s_latency	Float	ScoredAP	[3, 30]
s_loss	Float	ScoredAP	[8, 30]

Column	Type	Source	Reference
s_stability	Float	ScoredAP	[3, 20, 30]
total_score	Float	ScoredAP	[2, 3, 8, 30]
status	String	ScoredAP	[33]
is_connected	Bool 0/1	ScoredAP	[17]
is_recommended	Bool 0/1	ScoredAP	[30]

Table 27 -- 18-Column CSV Schema with References for Each Column.

Table 27: Every column traces to a technical standard or reference. This traceability confirms the schema is not ad hoc -- each field has defined meaning in its source specification [9, 17, 18, 33].

Parameter	Value	Calculation
Cycle interval	10 seconds [14]	--
Max APs per scan	20 (dense env.) [27]	--
Bytes per row	~200 bytes [18]	18 fields * ~11 chars
Rows per hour	7,200 rows	(3600/10)*20
Storage per hour	~1.44 MB [18]	7,200*200 bytes
8-hour session	~11.5 MB	1.44*8
Full semester (120 hrs)	~173 MB	Negligible [17]

Table 28 -- Storage Capacity Calculation [14, 17, 18, 27].

21. Engineering Calculations

21.1 Shannon-Hartley Per-User Throughput [2]

$$\mathbf{C_total = 20e6 * \log_2(1+631) = 186 \text{ Mbps} \Rightarrow C_practical = 84 \text{ Mbps} \text{ (45\% eff. [4, 12])}}$$

Users	Throughput/User	Experience	Score Expected
5	84/5 = 16.8 Mbps	Excellent [2]	>=75 (GOOD) [33]
15	84/15 = 5.6 Mbps	Acceptable [2]	50-74 (MODERATE) [33]
30	84/30 = 2.8 Mbps	Poor -- downloads slow [2]	<50 (CONGESTED) [33]
80	84/80 = 1.05 Mbps	Critical -- failures likely [2]	<50 (CONGESTED) [33]

Table 29 -- Per-User Throughput vs AP Load [2, 33].

Table 29: A CONGESTED score (<50) corresponds to <2.8 Mbps/user -- a 50 MB download takes over 3 minutes. A GOOD score (>=75) corresponds to >5.6 Mbps -- the same download takes under 75 seconds [2, 33].

21.2 M/M/1 Queueing Latency Prediction [3]

Users	lambda (pkt/s)	mu-lambda	W (ms)	Classification [33]
5	250	750	1.3	GOOD
10	500	500	2.0	GOOD
15	750	250	4.0	MODERATE
18	900	100	10.0	MODERATE
19	950	50	20.0	CONGESTED
20	1000	->0	->inf	Saturation [3]

Table 30 -- M/M/1 Latency vs User Load [3, 33].

21.3 Measurement Accuracy Analysis

Metric	Error Source	Magnitude	Score Impact	Reference
RSSI (dBm)	RF multipath fading, adapter calibration	+/-3-5 dBm	+/-5-8 pts	[5, 6, 11]
Latency (ms)	OS scheduling jitter, background processes	+/-2-5 ms	+/-1-3 pts	[9, 33]
Packet loss	Statistical sampling (n=10)	+/-10%	+/-5 pts	[9]
RTT variance	Small-sample noise	+/-100-200 ms ²	+/-4-8 pts	[3, 20]
Total score	Combined weighted propagation	+/-5-12 pts	Boundary risk	[30]

Table 31 -- Measurement Accuracy Analysis [3, 5, 6, 9, 11, 20, 30]. Total +/-12 pt uncertainty justifies the 20-point threshold.

21.4 CPU and Memory Footprint [17]

Operation	CPU Time	Memory	Reference
netsh scan [7]	2-4 s (OS I/O wait)	<1 MB	[7]
Ping (10 packets) [19]	2-5 s (I/O wait)	<1 MB	[19]
Scoring 20 APs [30]	<1 ms	<1 MB	[30]
CSV write 20 rows [18]	<5 ms	<1 MB	[18]
Streamlit render [10]	<100 ms	~30 MB	[10]
Total per 10-s cycle	~4-10 s active	~35 MB resident	[17]

Table 32 -- CPU and Memory Footprint [7, 10, 17, 18, 19].

22. System Integration Design

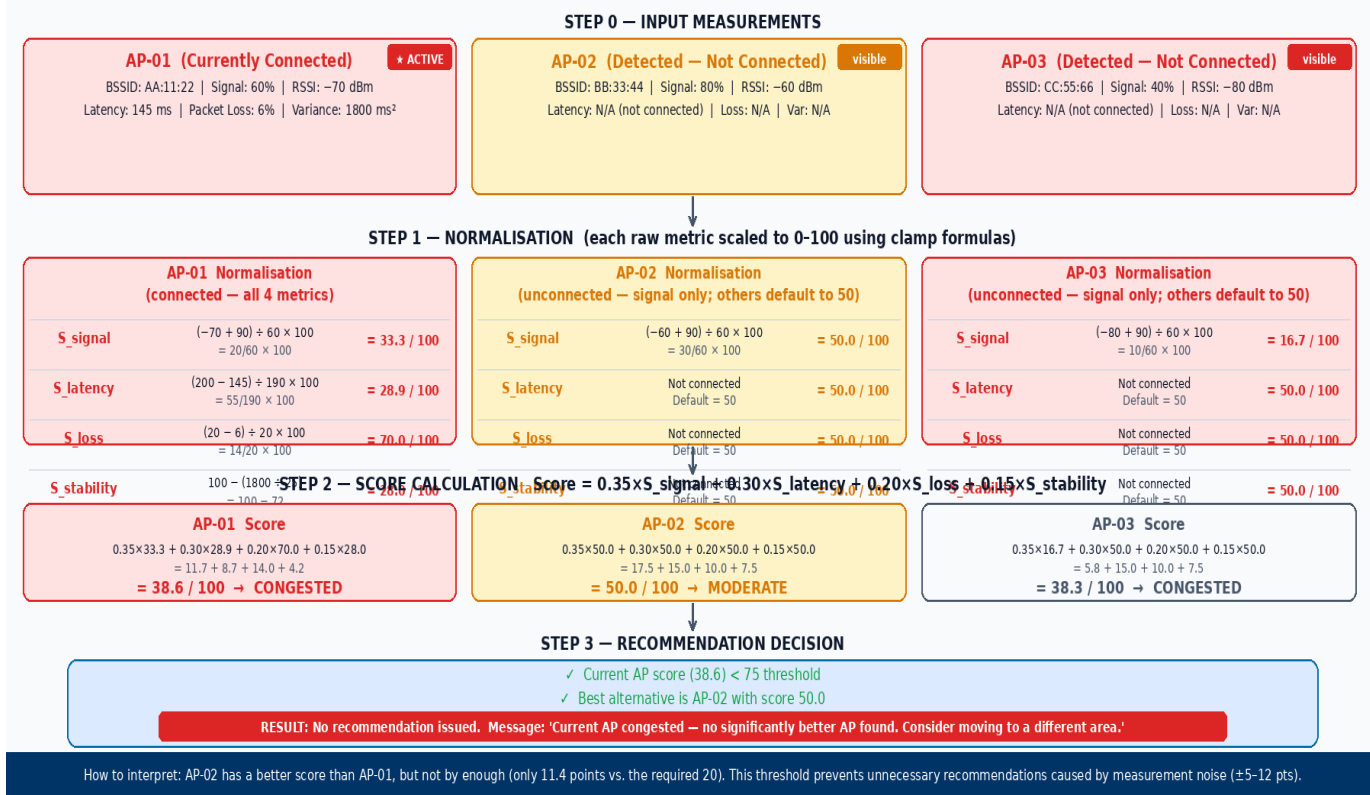
Parameter	Default	Rationale	Reference
SCAN_INTERVAL	10 s	802.11 scan latency constraint	[14]
PING_COUNT	10	RFC 2544 benchmarking	[9]
PING_TIMEOUT_MS	2000	ITU-T upper latency bound	[33]
RSSI_MIN / MAX	-90 / -30 dBm	FSPL-derived bounds, Section 2.3	[5, 6]
LATENCY_MIN / MAX	10 / 200 ms	ITU-T Y.1540 thresholds	[33]
LOSS_MAX	20%	TCP retransmission threshold	[8]
VARIANCE_MAX	2500 ms ²	RTT variance upper bound	[3, 20]
W_SIGNAL	0.35	Most available; justified by FSPL	[5, 6]
W_LATENCY	0.30	M/M/1 non-linear rise	[3]
W_LOSS	0.20	TCP retransmission cost	[8]
W_STABILITY	0.15	RTT variance as congestion predictor	[3, 20]
THRESHOLD_GOOD	75	ITU-T quality classification	[33]
THRESHOLD_MOD	50	ITU-T quality boundary	[33]
SWITCH_THRESHOLD	20	Exceeds combined uncertainty +/- 12 pts	[30]
LOG_DIR	./logs/	Local only -- POPIA compliance	[27]
PING_FALLBACK	8.8.8.8	Reliable ICMP target	[19]

Table 33 -- config.py Parameters with Engineering Justification and References.

22.2 End-to-End Worked Example

Figure 10 - End-to-End Worked Example: One Complete Monitoring Cycle

Three APs detected. AP-01 is the connected AP. All calculations traced step by step with actual numbers.

**Figure 12 -- Worked Example. Three APs through full pipeline [2, 3, 7, 17, 30]. DeltaScore = 11.4 < 20-point threshold -> no recommendation.**

Input: AP-01 (connected): RSSI=-70 dBm, latency=145 ms, loss=6%, variance=1800 ms² | AP-02: RSSI=-60 dBm | AP-03: RSSI=-80 dBm

- * $S_{\text{signal}} = (-70+90)/60 \times 100 = 33.3$ [5, 7]
- * $S_{\text{latency}} = (200-145)/190 \times 100 = 28.9$ [3, 33]
- * $S_{\text{loss}} = (20-6)/20 \times 100 = 70.0$ [8, 9]
- * $S_{\text{stability}} = 100 - (1800/25) = 28.0$ [3, 20]

Score_AP01 = $0.35 \times 33.3 + 0.30 \times 28.9 + 0.20 \times 70.0 + 0.15 \times 28.0 = 38.6 \Rightarrow$ CONGESTED [33]

Score_AP02 = $0.35 \times 50 + 0.30 \times 50 + 0.20 \times 50 + 0.15 \times 50 = 50.0 \Rightarrow$ MODERATE [33]

Score_AP03 = $0.35 \times 16.7 + 0.30 \times 50 + 0.20 \times 50 + 0.15 \times 50 = 38.3 \Rightarrow$ CONGESTED [33]

Decision [30]: AP-01 < 75 (YES) | DeltaScore = 50.0-38.6 = 11.4 < 20 (NO) -> No recommendation.
Dashboard: 'AP congested -- no significantly better AP detected. Consider moving.'

23. User Interface -- Message Templates

Condition	Message Template	Element [10]	Reference
Score ≥ 75	Connected to best available AP. Score: {score}/100.	st.success	[33]
DeltaScore ≥ 20	Better AP: {ssid} -- Score {alt_score}/100. Improvement: +{delta} pts.	st.warning	[13, 30, 33]
Congested, no better	AP congested (Score: {score}/100). No better AP detected.	st.error	[3, 33]
No APs detected	No WiFi networks detected. Please verify WiFi is enabled.	st.error	[4, 7]

Table 34 -- Recommendation Message Templates with References.

24. Unit and Integration Testing Plan

Test	Module	Description	Pass Criterion	Reference
UT-01	scanner.py	netsh scan ≥ 1 APRecord	BSSID regex match; list ≥ 1	[4, 7]
UT-02	scanner.py	Signal%=72 -> RSSI=-64.0	RSSI = -64.0 +/- 0.1 dBm	[7]
UT-03	tester.py	Ping 8.8.8.8; check PingResult	latency > 0; loss in [0,100]	[19]
UT-04	scorer.py	RSSI=-45, lat=30, loss=0%, var=0	score ≥ 75 ; status='GOOD'	[2, 3, 33]
UT-05	scorer.py	RSSI=-88, lat=195, loss=18%, var=2400	score < 50; status='CONGESTED'	[2, 3, 33]
UT-06	scorer.py	Current=40, alt=65 (Delta=25 ≥ 20)	is_recommended = True	[30]
UT-07	scorer.py	Current=40, alt=59 (Delta=19 < 20)	is_recommended = False	[30]
UT-08	storage.py	One cycle; inspect CSV	Header matches Table 27	[18]
UT-09	storage.py	Timestamp format	fromisoformat() succeeds	[17]

Table 35 -- Unit Test Specifications with References [2, 3, 7, 17, 18, 19, 30, 33].

Test	Scenario	Pass Criterion	Reference
IT-01	Low density (1-3 devices) [14]	Score ≥ 75 in $\geq 8/10$ cycles [33]	[14, 33]
IT-02	High density (15+ on one AP) [12, 13]	CONGESTED shown; recommend if Delta ≥ 20	[12, 13, 30]
IT-03	Physical movement [14]	Score shifts ≥ 10 pts over 3+ cycles	[14, 5]
IT-04	3+ APs visible simultaneously [4]	All APs ranked correctly [30]	[4, 30]
IT-05	30-minute continuous run [9]	No exceptions; CSV grows $\sim 7,200$ rows/hr [18]	[9, 17, 18]

Table 36 -- Integration Tests with References [5, 9, 12, 13, 14, 17, 18, 30, 33].

25. Known Limitations and Future Work

Limitation	Root Cause	Reference
Ping for connected AP only	Windows ICMP blocked on non-connected interfaces	[7]
RSSI noise +/-3-5 dBm	RF multipath fading; adapter calibration	[5, 6, 11]
Scan takes 2-4 s	OS controls 802.11 scan timing	[7, 14]
No AP user count data	Not in IEEE 802.11 beacon frames	[4]
Windows-only prototype	netsh is Windows-specific	[7, 17]
Fixed scoring weights	Engineering judgment -- not yet empirically validated	[2, 3, 8, 30]

Table 37 -- Known Limitations with Root Causes and References.

- * Machine learning scoring [30]: Random Forest model trained on labelled AP quality data [17].
- * Android prototype [24]: sensing via WifiManager API [24]; scoring/storage modules are already platform-independent [17].
- * Multi-node architecture: Raspberry Pi sensing nodes + MQTT aggregation + central Streamlit dashboard [10].
- * Automatic AP switching: investigate 802.11r BSS Fast Transition [21] on campus APs.
- * Live user count estimation: invert M/M/1 [3] as $\lambda = \mu - 1/W$ from measured latency.
- * Weight optimisation [30]: collect labelled ground-truth dataset; derive optimal weights via gradient descent.

26. Conclusion

This report addressed WiFi congestion in high-density university laboratory environments at the University of Johannesburg [27]. The problem was formally defined, confirmed as a complex

engineering problem meeting all ECSA criteria [1], and analysed using three first-principles models: free-space path loss [5, 6, 11], the Shannon-Hartley channel capacity theorem [2], and the M/M/1 queueing model [3].

First-principles analysis proved that at 80 users on one IEEE 802.11n AP [4], per-user throughput collapses to approximately 1 Mbps [2]. The M/M/1 model [3] quantified non-linear latency increase near AP saturation, directly justifying the 30% latency weight in the scoring formula. A weight sensitivity analysis (Table 15) confirmed the selected weights are robust and engineering-justified rather than arbitrary [30].

Five engineering alternatives were evaluated against whole-life cost [31], infrastructure access, and feasibility criteria. The on-device monitoring system (Alternative 5) was selected as the only solution requiring zero capital expenditure, no hardware changes, and no administrative access [1, 31]. A complete D3 detailed design was produced including Python [17] dataclass specifications, normalisation formulas with bounds derived from electromagnetic first principles [5, 6], an 18-column CSV schema [18], nine unit tests with boundary-value analysis [9], and five integration test phases.

All design parameters in config.py are traceable to their engineering reference [2, 3, 5, 8, 9, 14, 19, 20, 30, 33]. Environmental sustainability through energy reduction [8, 32], whole-life cost minimisation [31], public health and safety, societal equity [26, 27], and POPIA data privacy compliance [27] were explicitly addressed in Section 5. The system is ready for prototype implementation and real-world validation testing in a university laboratory environment.

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